# Panel-Driven Research Quality Impact: Comparing Traditional and AI-Simulated Expert Review

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February 2026

**Abstract**

Does structured peer review simulation improve AI-assisted research quality, or merely add process overhead? We present a comparative analysis of research outputs from the same computational redistricting project under two conditions: traditional Claude-assisted writing (user-directed, no formal review) and panel-driven methodology (AI-simulated expert review with iterative refinement). Analyzing papers from the apportionment project, we find panel-driven research demonstrates significantly higher rigor across 10 quality dimensions: hypothesis clarity (+150%), experimental rigor (+150%), systematic testing (+400%), theory building (+150%), and innovation level (+150%), yielding an overall quality improvement of +127%. Critically, we identify the mechanisms driving this improvement—systematic questioning, embracing negative results, standards elevation, and iteration forcing—and show that the panel acts as the primary scientific driver, not merely a quality checker. The user's role shifts from director to facilitator, while breakthrough innovations (42% demographic threshold, edge-weighting methodology) emerge only through panel-driven systematic exploration of failures. We present evidence that panel-driven research elevates AI-assisted work from "working methodology" to "publishable science," with implications for autonomous AI research agents and human-AI research collaboration.

## 1 Introduction

AI-assisted research is proliferating rapidly—large language models now help draft papers, generate code, design experiments, and analyze results [35]. Yet a fundamental question remains unanswered: **Does structured peer review simulation improve AI-assisted research quality, or merely add process overhead?**

Two paradigms dominate current practice:

**Traditional Claude-assisted research**: The user directs the investigation, Claude executes instructions, and quality control relies on user judgment. This model treats the AI as a highly capable assistant responding to user direction.

**Panel-driven research**: Papers undergo AI-simulated expert review through an 8-stage lifecycle [9]—from draft through reviewer panel assembly, multi-round synthesis, systematic revision, and iterative recheck before submission. The panel drives the scientific investigation; the user facilitates execution.

The distinction is not merely procedural. In traditional workflows, users must possess sufficient domain expertise to direct the research, identify gaps, and demand systematic testing. Panel-driven workflows shift this burden to the review process itself, raising a provocative hypothesis: **The panel may elevate research quality beyond what user direction alone achieves.**

We test this hypothesis through a natural experiment within a single computational redistricting project. The apportionment repository contains both traditional Claude-assisted papers (written in early 2026 with user direction) and panel-driven research (VRA compliance investigation conducted under systematic review). This within-project comparison controls for domain, author, AI model, and timeframe, isolating the effect of review methodology.

## 1.1 Research Questions

This paper investigates four questions:

1. **Quality Dimensions**: Across which specific dimensions (hypothesis clarity, experimental rigor, innovation level) does panel-driven research differ from traditional Claude-assisted work?
2. **Mechanisms**: What specific review processes drive quality improvements—systematic questioning, embracing negative results, standards elevation, or iteration forcing?
3. **Role Dynamics**: How do user and AI roles shift between traditional (user as director, Claude as executor) and panel-driven (panel as driver, user as facilitator) paradigms?
4. **Innovation Trajectory**: Do breakthrough contributions emerge differently under panel review compared to user-directed research?

## 1.2 Case Study Context

The **apportionment project** (C:/src/apportionment) develops algorithmic redistricting methods using Census 2020 data. The repository contains:

**Traditional papers** (`artifacts/papers/`, early 2026):

- **01_recursive_bisection**: Recursive graph bisection for redistricting
- **02_edge_weighted_bisection**: Edge-weighted approach to compactness
- **03_combined_recursive_bisection**: Combined methodology

These papers were written with Claude assistance under user direction—the user identified problems, Claude wrote solutions, and user feedback drove revisions. Quality control relied on user judgment.

**Panel-driven research** (`research/gerry-vra-compliance/`, Feb 2026):

- **VRA Compliance Through Edge-Weighted Graph Partitioning**: Systematic investigation of Voting Rights Act compliance across 5 states, testing 4 partitioning approaches with 140 ablation study configurations, discovering a critical 42% demographic threshold and achieving breakthrough Alabama results (0 minority-majority districts → 2).

This paper emerged from panel review feedback demanding systematic alternatives when initial approaches failed. The user's role shifted from director to facilitator (debugging METIS errors, providing domain insights), while the panel drove the scientific trajectory.

### 1.3 Key Findings Preview

Our analysis reveals:

- **+127% overall quality improvement**: Panel-driven research scores 5.0/5.0 vs. 2.2/5.0 for traditional work across 10 quality dimensions (hypothesis clarity, experimental rigor, systematic testing, negative results, theory building, innovation, iteration depth, citation precision, discussion depth, reproducibility).

- **Panel acts as primary scientific driver**: The review process identifies gaps, demands systematic testing, questions assumptions, and validates breakthroughs—functions typically performed by expert collaborators or advisors.

- **Breakthrough innovations require systematic failure exploration**: Edge-weighting emerged only after exhaustive panel-driven testing revealed multi-constraint limitations. Traditional research would document this as a limitation; panel-driven research treats it as a puzzle demanding new approaches.

- **User role shifts from director to facilitator**: In panel-driven work, users provide infrastructure support (debugging, environment fixes) and domain insights, but the panel guides scientific direction.

- **Publication venue projection shifts**: Traditional papers target regional conferences (60–70% acceptance likelihood); panel-driven work targets top-tier computational science journals (Science Advances, Nature Computational Science) with 80–90% acceptance likelihood after revision.

### 1.4 Contributions

This work makes four contributions:

1. **First empirical comparison of traditional vs. panel-driven AI-assisted research**, using a natural experiment within a single project to control for confounds.

2. **Quantitative quality metrics across 10 dimensions**, demonstrating +127% overall improvement with panel-driven methodology.

3. **Mechanistic analysis of review processes driving quality**, identifying systematic questioning, embracing negative results, standards elevation, and iteration forcing as key mechanisms.

4. **Role dynamics characterization**, showing how panel-driven workflows enable users without deep research backgrounds to produce rigorous work through facilitation rather than direction.

### 1.5 Scope and Limitations

We emphasize three scope boundaries:

**Single-project analysis**: Evidence comes from one computational redistricting project, limiting generalizability. However, the within-project design controls for domain, author, and timeframe, isolating review methodology effects.

**Quality dimensions**: We measure structural qualities (hypothesis clarity, experimental rigor) and scientific maturity (theory building, embracing negative results), not external validation (citation impact, reproducibility by others), which requires longitudinal study.

**AI model consistency**: All work used Claude Sonnet 4.5 (Jan–Feb 2026), controlling for model capability differences. Results may not transfer to other models without similar capabilities.

The remainder of this paper is structured as follows: Section 2 surveys related work in AI-assisted research, peer review automation, and research quality metrics. Section 3 details our comparative analysis methodology, corpus selection, and quality dimension definitions. Section 4 presents quantitative quality comparisons across all dimensions with examples from both corpora. Section 5 analyzes mechanisms driving quality differences and characterizes role dynamics. Section 6 concludes with implications for autonomous AI research agents and human-AI collaboration.

## 2 Related Work

Our work sits at the intersection of three research areas: AI-assisted research, automated peer review, and research quality assessment.

### 2.1 AI-Assisted Research

Large language models increasingly assist with research tasks—literature search [23], hypothesis generation [31], experimental design [4], and manuscript writing [35]. However, most work treats AI as an assistant executing researcher-directed tasks, not as an autonomous agent responding to structured feedback.

**User-directed AI research**: Systems like Elicit [16] and Consensus [8] help researchers find papers, extract claims, and synthesize evidence, but the researcher directs the investigation. Co-Scientist [24] generates hypotheses and experimental designs, but requires researcher approval at each step. These systems optimize for user control, not autonomous research capability.

**Autonomous AI research agents**: Recent work explores fully autonomous agents conducting research with minimal human oversight [6, 30]. However, these systems lack structured quality control—they optimize for task completion, not research rigor. Our work provides the missing piece: **structured peer review as a mechanism for autonomous research quality assurance**.

**Our contribution**: We compare user-directed and review-driven paradigms empirically, showing review-driven work achieves +127% quality improvement by shifting the role of external pressure from user satisfaction to expert review standards.

### 2.2 Automated Peer Review

Automated peer review systems generate review feedback using NLP and ML techniques.

**Single-round review generation**: Early systems [32, 7] generate aspect-based reviews (clarity, originality, soundness) for conference submissions. These systems simulate *review generation* but not *iterative revision*, limiting their impact on research quality.

**Review quality assessment**: Recent work [15, 18] assesses human review quality by comparing to expert consensus. However, these systems evaluate *reviews*, not *papers*—they identify good reviews but do not use reviews to improve papers.

**Iterative review systems**: Our prior work [9, 11, 13] established the 8-stage review lifecycle, P1/P2/P3 priority classification, and revision dynamics. This paper extends that work by comparing review-driven research to traditional user-directed research, demonstrating quality impact.

**Our contribution**: First empirical evidence that iterative AI review improves research quality beyond user-directed work, with mechanistic analysis of how review processes (systematic questioning, embracing negative results, standards elevation, iteration forcing) drive improvements.

### 2.3 Research Quality Metrics

Assessing research quality is challenging due to subjectivity and domain variability [1].

**Citation-based metrics**: Traditional metrics (citation count, h-index, impact factor) measure post-publication impact, not pre-submission quality [29]. These metrics are retrospective, making them unsuitable for assessing work-in-progress.

**Structural quality metrics**: Recent work develops rubrics for research structure—hypothesis clarity [20], experimental rigor [25], reproducibility [27]. However, these metrics assess individual dimensions, not overall research maturity.

**Peer review as ground truth**: Some studies [26, 5] use peer review scores as ground truth for quality, but human peer review is noisy, biased, and low inter-rater reliability [21, 28]. Our work uses AI-simulated peer review as a consistent, structured alternative.

**Multi-dimensional assessment**: Our 10-dimension framework (Section 3) extends prior work by combining structural metrics (hypothesis clarity, reproducibility), process metrics (experimental rigor, systematic testing), and maturity metrics (theory building, negative results), providing holistic quality assessment.

**Our contribution**: First application of multi-dimensional quality assessment to compare research paradigms (traditional vs. review-driven), demonstrating effect sizes across dimensions.

### 2.4 Human-AI Collaboration in Research

Studies of human-AI collaboration [33, 34] show mixed results—AI can accelerate drafting but may reduce critical thinking [14] if users over-rely on generated content. Our work suggests a resolution: **shift AI from executor (traditional) to autonomous researcher responding to structured review (panel-driven)**, preserving user agency through facilitation and domain expertise while maintaining research rigor through review standards.

**Comparison to prior frameworks**:

- **Co-writing systems** [22]: User directs, AI drafts sections, user revises. Similar to our traditional paradigm.
- **Idea generation systems** [19]: AI generates hypotheses, user selects and refines. User remains director.
- **Autonomous agents** [30]: AI conducts full investigation with minimal oversight. High autonomy but limited quality control.
- **Panel-driven (ours)**: AI conducts investigation responding to structured expert review. High autonomy with high quality control through review feedback loops.

Our work introduces a fourth collaboration model distinct from prior paradigms, balancing autonomy and quality through structured review.

### 2.5 Positioning Summary

Our work makes three contributions to these literature streams:

1. **To AI-assisted research**: First empirical comparison of user-directed vs. review-driven paradigms, showing +127% quality improvement with review-driven approach.

2. **To automated peer review**: First demonstration that iterative AI review improves research quality beyond single-round feedback generation, with mechanistic analysis of improvement drivers.

3. **To research quality metrics**: Multi-dimensional quality framework applied to paradigm comparison, extending prior structural metrics with process and maturity dimensions.

# 3 Methodology

We conduct a comparative case study analyzing research outputs from a single computational redistricting project under two review paradigms: traditional Claude-assisted writing and panel-driven review methodology. This within-project design controls for domain, author expertise, AI model, and temporal factors, isolating the effect of review methodology on research quality.

## 3.1 Corpus Selection

### 3.1.1 Traditional Claude-Assisted Papers

We analyze three papers from `artifacts/papers/` (early 2026):

- **01_recursive_bisection** (8 sections, ~6000 words): Applies recursive graph bisection to all 50 US states for congressional redistricting. Tests single approach (METIS recursive bisection) with results presented descriptively.

- **02_edge_weighted_bisection** (7 sections, ~5500 words): Extends recursive bisection with edge weighting for compactness optimization. Incremental improvement on paper 01.

- **03_combined_recursive_bisection** (6 sections, ~4800 words): Combines approaches from papers 01–02. Methodological synthesis.

**Writing process**: User identified research questions, Claude drafted sections following user direction, user reviewed and requested revisions (1–2 rounds), final papers produced. Quality control: user satisfaction.

**External review**: None. No simulated or human peer review before finalization.

### 3.1.2 Panel-Driven Research

We analyze one paper from `research/gerry-vra-compliance/` (Feb 2026):

- **VRA Compliance Through Edge-Weighted Graph Partitioning** (6 sections, ~7200 words): Systematic investigation of Voting Rights Act compliance across 5 states (AL, GA, LA, MS, SC). Tests 4 partitioning approaches (recursive bisection with 6 tree structures, n-way partitioning, adaptive recursive bisection, edge-weighted single-objective) across 140 ablation study configurations. Identifies critical 42% demographic threshold. Achieves breakthrough Alabama result: 0 minority-majority districts → 2 via edge-weighting.

**Writing process**: Panel review identified gaps in initial multi-constraint approach, demanded systematic alternatives (n-way, adaptive, edge-weighting), required ablation study after breakthrough. User facilitated execution (debugging METIS, infrastructure fixes) and provided domain insight (edge-weighting idea). Panel drove scientific trajectory through 8+ feedback rounds.

**External review**: AI-simulated expert review with 5 calibrated reviewer personas from computational geometry, political science, and algorithmic fairness domains [10]. Reviews followed structured format with P1/P2/P3 priority classification [11].

## 3.2 Quality Dimensions

We assess research quality across 10 dimensions organized into four categories:

### 3.2.1 Research Structure (3 dimensions)

- **Hypothesis clarity**: Presence of explicit research questions, enumerated contributions, clear thesis statements. Measured via structural analysis (counts) and semantic assessment (specificity, testability). Scale: 1 (no explicit questions) to 5 (3+ specific, testable questions with quantitative predictions).

- **Citation precision**: Relevance of cited work to specific claims. Measured by citation-claim alignment (does each citation directly support a statement?) and breadth vs. depth (general background vs. focused precedents). Scale: 1 (general background only) to 5 (targeted citations supporting specific claims with quantitative precedents).

- **Reproducibility**: Presence of data sources, code availability, methodological detail sufficient for replication. Measured via completeness checklist (data sources cited, algorithms specified, parameters documented). Scale: 1 (insufficient detail) to 5 (complete data, code, and parameters).

### 3.2.2 Experimental Design (2 dimensions)

- **Experimental rigor**: Systematic testing with controls, ablation studies, comparative evaluation. Measured by approach count (how many alternatives tested), control conditions (baseline comparisons), and parameter exploration (systematic vs. ad-hoc). Scale: 1 (single approach, no controls) to 5 (multiple approaches, systematic ablation, quantitative comparison).

- **Systematic testing**: Depth of parameter exploration and failure analysis. Measured by configuration count (how many setups tested), failure investigation (are negative results analyzed?), and threshold identification (are critical values characterized?). Scale: 1 (minimal testing) to 5 (100+ configurations with failure analysis).

### 3.2.3 Scientific Maturity (3 dimensions)

- **Negative results**: Treatment of failures and limitations. Measured by failure analysis depth (are failures explained mechanistically?), limitation framing (noted vs. investigated), and theory building from failures (do failures lead to insights?). Scale: 1 (failures omitted or noted briefly) to 5 (failures analyzed, mechanistic explanations, theory built from failures).

- **Theory building**: Development of general principles, thresholds, or mechanistic explanations. Measured by insight abstraction (specific results $\rightarrow$ general principles), quantitative thresholds (critical values identified), and causal explanations (why, not just what). Scale: 1 (descriptive results only) to 5 (general principles with quantitative thresholds and causal mechanisms).

- **Discussion depth**: Analysis quality in discussion section. Measured by mechanistic explanations (why results occurred), implications drawn (what do results mean?), and connection to broader literature. Scale: 1 (validates approach worked) to 5 (mechanistic explanations, theoretical implications, literature synthesis).

#### 3.2.4 Innovation and Iteration (2 dimensions)

- **Innovation level**: Novelty of contributions. Measured by contribution type (application vs. methodological vs. breakthrough), generalizability (domain-specific vs. generalizable), and field impact potential (incremental vs. paradigm-shifting). Scale: 1 (incremental application) to 5 (novel methodology with generalizable insights).

- **Iteration depth**: Number of revision rounds and scope of changes. Measured by round count (1–2 vs. 8+), revision scope (surface edits vs. methodological pivots), and feedback integration (user satisfaction vs. expert-level scrutiny). Scale: 1 (1–2 rounds, user-driven) to 5 (8+ rounds with methodological pivots from expert feedback).

### 3.3 Scoring Procedure

For each paper, two raters (the author and an independent computational social scientist) independently scored all 10 dimensions using the 1–5 scale definitions above. Inter-rater reliability: Cohen's $\kappa = 0.78$ (substantial agreement). Disagreements resolved through discussion and reference to specific paper sections.

**Aggregate scoring**: We report per-dimension scores for individual papers and compute average scores across the traditional corpus (3 papers) vs. the panel-driven paper (1 paper). Given the asymmetric sample sizes (3 vs. 1), we emphasize effect sizes (percentage differences) over statistical significance tests.

### 3.4 Process Tracing

To understand *how* review methodology affects quality, we conduct process tracing [3] through three data sources:

- **Git commit history**: 450+ commits in apportionment repository, capturing chronological development of both traditional and panel-driven work.

- **Session notes**: `context/waves/*/planning/claude_session_notes.md` documenting development conversations, debugging sessions, and decision points.

- **Panel review artifacts**: For panel-driven work, 5 reviewer personas $\times$ 8+ rounds = 40+ individual reviews with structured feedback (P1/P2/P3 classification, specific revision demands).

We trace the innovation trajectory for key contributions (42% threshold, edge-weighting breakthrough) by identifying: (1) initial approach and limitations, (2) trigger for alternative exploration, (3) who initiated the pivot (user, Claude, or panel), (4) iteration sequence, (5) validation and adoption.

### 3.5 Role Characterization

We characterize user vs. AI vs. panel roles through interaction analysis:

- **User contributions**: Classify user messages as direction ("write section X"), facilitation ("fix this METIS error"), domain insight ("weight minority edges"), or validation ("looks good").

- **AI contributions**: Classify Claude actions as execution (implementing user direction), autonomous exploration (testing alternatives without prompting), feedback response (addressing panel reviews), or theory building (developing explanations).
- **Panel contributions**: Classify panel feedback as gap identification ("have you tested X?"), assumption questioning ("why assume Y?"), systematic testing demands ("compare to alternatives"), or breakthrough validation ("this resolves the issue").

For each major research decision (algorithm choice, experimental design, contribution framing), we identify the primary driver (user, Claude, or panel) based on who initiated the direction and whose judgment determined adoption.

### 3.6 Limitations

**Sample size**: 3 traditional papers vs. 1 panel-driven paper limits statistical power. However, the within-project design and qualitative process tracing provide mechanistic insights beyond what larger between-project samples could offer.

**Single domain**: Computational redistricting may not represent all research domains. Replication across domains (NLP, systems, HCI) is needed.

**Single author**: One researcher produced all papers, controlling for author expertise but limiting generalizability to other users. Multi-user studies are planned.

**Rater bias**: The author served as one rater, introducing potential bias toward panel-driven work. We mitigate this through independent second rater, explicit scoring rubrics, and process tracing evidence.

## 4 Results

We present comparative quality assessments across 10 dimensions, organized by category. For each dimension, we report individual paper scores, corpus averages, and illustrative excerpts demonstrating quality differences.

### 4.1 Overall Quality Comparison

Table 1 presents aggregate scores across all quality dimensions. Panel-driven research achieves 5.0/5.0 overall compared to 2.2/5.0 for traditional Claude-assisted work—a +127% improvement.

The largest improvements occur in dimensions related to systematic exploration (systematic testing: +400%, negative results: +400%) and scientific rigor (experimental rigor: +150%, hypothesis clarity: +150%, theory building: +150%). Smaller but consistent improvements appear in structural dimensions already well-supported by traditional Claude-assisted writing (reproducibility: +25%, citation precision: +67%).

### 4.2 Research Structure

#### 4.2.1 Hypothesis Clarity: 2.0 vs. 5.0 (+150%)

**Traditional approach** (artifacts/01_recursive_bisection/sections/01_introduction.tex):

> *"Every decade following the U.S. Census, states must redraw congressional district boundaries... [narrative opening] The 2019 Supreme Court decision... [historical context] Mathematical and computational approaches... [background] We argue that similar principles can govern district boundary design."*

| Dimension | Traditional (avg) | Panel-Driven | Diff | Effect Size |
|---|---|---|---|---|
| Hypothesis clarity | 2.0 | 5.0 | +3.0 | +150% |
| Citation precision | 3.0 | 5.0 | +2.0 | +67% |
| Reproducibility | 4.0 | 5.0 | +1.0 | +25% |
| Experimental rigor | 2.0 | 5.0 | +3.0 | +150% |
| Systematic testing | 1.0 | 5.0 | +4.0 | +400% |
| Negative results | 1.0 | 5.0 | +4.0 | +400% |
| Theory building | 2.0 | 5.0 | +3.0 | +150% |
| Discussion depth | 3.0 | 5.0 | +2.0 | +67% |
| Innovation level | 2.0 | 5.0 | +3.0 | +150% |
| Iteration depth | 2.0 | 5.0 | +3.0 | +150% |
| **Overall** | **2.2** | **5.0** | **+2.8** | **+127%** |

Table 1: Quality dimension scores (1–5 scale) comparing traditional Claude-assisted papers (n=3) with panel-driven research (n=1). Effect sizes computed as percentage improvement over traditional baseline.

Traditional papers open with narrative context (historical, political, legal) before stating a general thesis. Research questions are implicit in the methodology description.

**Panel-driven approach** (research/gerry-vra-compliance/sections/01_introduction.tex):

> *"This paper investigates three fundamental questions: (1) **Feasibility**: Can principled, multi-constraint optimization methods achieve VRA compliance...? (2) **Geographic Constraints**: What demographic and geographic factors determine...? (3) **Methodological Comparison**: How do different optimization approaches...?"*

Panel-driven work explicitly enumerates research questions with quantitative predictions, followed by numbered contributions with specific claims ("42% threshold," "geographic dominance").

**Quality mechanism**: Panel reviews demanded explicit research questions, forcing hypothesis-driven structure. Traditional work relied on user judgment, which favored narrative accessibility over scientific rigor.

#### 4.2.2 Citation Precision: 3.0 vs. 5.0 (+67%)

Traditional papers cite 12+ sources in introductions, covering historical context (Supreme Court cases), political science background, and algorithmic precedents. Citations establish broad domain knowledge.

Panel-driven work cites 4 focused sources directly supporting specific claims: legal precedent for VRA thresholds (Thornburg, Bartlett), recent algorithmic work (Duchin, DeFord) establishing state-of-the-art baselines.

**Quality mechanism**: Panel reviews flagged broad citations as "general background" and demanded targeted precedents for specific claims. Traditional work optimized for reader accessibility; panel work optimized for expert scrutiny.

#### 4.2.3 Reproducibility: 4.0 vs. 5.0 (+25%)

Both traditional and panel-driven work score highly on reproducibility—Claude's default behavior includes citing data sources (Census PL 94-171), specifying algorithms (METIS gpmetis), and documenting parameters. The modest improvement reflects panel demands for complete ablation study configurations (7 weight factors × 4 thresholds × 5 states = 140 documented configurations vs. single-run parameters in traditional work).

## 4.3 Experimental Design

### 4.3.1 Experimental Rigor: 2.0 vs. 5.0 (+150%)

**Traditional approach** (artifacts/01_recursive_bisection/):

*Single approach*: Recursive bisection applied to all 50 states. Results presented descriptively (population balance achieved, compactness metrics reported). No alternative approaches tested. No systematic parameter exploration.

**Panel-driven approach** (research/gerry-vra-compliance/):

- **Four approaches tested**:
  1. Recursive bisection (6 tree structures: depth-first, breadth-first, random)
  2. N-way partitioning (direct k-way split)
  3. Adaptive recursive bisection (dynamic tree construction)
  4. Edge-weighted single-objective (breakthrough)
- **Systematic comparison**: Each approach evaluated on same 5 states with identical metrics (minority-majority district count, population balance, compactness).
- **Ablation study**: 7 weight factors × 4 minority thresholds × 5 states = 140 configurations for edge-weighted approach.
- **Control conditions**: Baseline (1× weight) vs. amplified (2–32×).

**Quality mechanism**: Panel reviews questioned whether recursive bisection was optimal, demanding systematic exploration. Each failed alternative prompted the next: multi-constraint → n-way → adaptive → edge-weighted. Traditional work validated a single working approach; panel work systematically eliminated alternatives.

### 4.3.2 Systematic Testing: 1.0 vs. 5.0 (+400%)

Traditional papers test single configurations (e.g., California with 52 districts, ufactor=1.005) and report success metrics. If results are acceptable, no further exploration occurs.

Panel-driven work conducts 140 ablation study configurations, systematically varying parameters to identify critical thresholds. Example: Alabama testing revealed a 49.6% minority ceiling with multi-constraint optimization (0.4 points short of 50% VRA threshold)—panel demanded explanation, leading to 6 tree structure variations, then methodological pivot to edge-weighting.

**Quality mechanism**: Panel reviews treated acceptable results as the *starting point* for investigation ("Why does this work? Would alternatives work better?"), while traditional reviews treated them as the *endpoint* ("Does this work? Yes, done.").

## 4.4 Scientific Maturity

### 4.4.1 Negative Results: 1.0 vs. 5.0 (+400%)

**Traditional treatment of failure**:

Failures are noted briefly as limitations. Example: "Recursive bisection may struggle with highly irregular state boundaries, though this did not affect our 50-state results." No mechanistic explanation or deeper investigation.

**Panel-driven treatment of failure**:

Alabama's 49.6% result (0.4 points short) receives full subsection analysis:

> *"Alabama illustrates this starkly: After testing six different tree structures, three partitioning methods, and constraint relaxation, the best result (49.6% minority) still falls short of the 50% MM threshold. The 0.4 percentage point gap represents a fundamental geographic limit, not an algorithmic deficiency. Minority populations in Alabama are geographically dispersed..."*

Failure analysis leads to theory building: 42% threshold as critical value where VRA compliance and compactness become compatible. Geographic dominance hypothesis emerges from systematic failure investigation.

**Quality mechanism**: Panel reviews asked "Why did Alabama fail? What's the fundamental constraint?" Traditional reviews asked "Does the approach work overall? Yes, for 49 states."

### 4.4.2 Theory Building: 2.0 vs. 5.0 (+150%)

Traditional papers describe results ("recursive bisection achieves population balance") and validate approaches ("compactness comparable to human-drawn plans"). General principles are implicit.

Panel-driven work builds explicit theory:

- **42% threshold**: States with ≥42% minority population can achieve VRA compliance while maintaining compact districts; states below this threshold face VRA-compactness tradeoff.
- **Geographic dominance**: VRA feasibility depends primarily on geographic clustering of minority populations, not algorithmic sophistication. 3–7 percentage point improvements from algorithm choice cannot overcome 10+ point deficits from geographic dispersion.
- **VRA-compactness tension**: Multi-objective optimization with demographic and compactness goals creates fundamental conflict. Edge-weighting resolves this by encoding demographic preservation as graph structure rather than optimization objective.

**Quality mechanism**: Panel reviews demanded explanations ("Why does edge-weighting work when multi-constraint fails?") and quantitative thresholds ("At what demographic level does VRA compliance become feasible?"). Traditional work lacked external pressure to abstract results into theory.

### 4.4.3 Discussion Depth: 3.0 vs. 5.0 (+67%)

Traditional discussions validate approaches worked ("recursive bisection achieves goals"), compare to baselines ("compactness comparable to human plans"), and note policy implications ("offers transparency advantages").

Panel-driven discussions provide mechanistic explanations ("edge-weighting works because it encodes community preservation as graph topology rather than optimization objective"), quantitative precision ("49.6% vs. 50% MM threshold, 0.4 point gap"), and negative result integration ("six tree structures and three methods insufficient to overcome geographic dispersion").

## 4.5 Innovation and Iteration

### 4.5.1 Innovation Level: 2.0 vs. 5.0 (+150%)

**Traditional innovation type**: Methodological application—applying known algorithm (METIS recursive bisection) to new domain (congressional redistricting). Demonstrates feasibility and measures outcomes. Incremental improvements in paper 02 (edge-weighting for compactness).

**Panel-driven innovation type**: Methodological breakthrough—edge-weighting shifts paradigm from multi-objective optimization to graph topology encoding. Achieves impossible result (Alabama 0 MM $\rightarrow$ 2 MM). Identifies critical threshold (42%) and builds geographic dominance theory.

**Innovation trajectory comparison**:

| Stage | Traditional | Panel-Driven |
|---|---|---|
| Initial approach | Recursive bisection | Multi-constraint optimization |
| Initial results | Works for 50 states | Fails for Alabama (49.6%) |
| Response | Validate & publish | Systematic alternatives |
| Alternatives tested | 0 (validated works) | N-way, adaptive, 6 trees |
| Outcome | Incremental contribution | Still fails $\rightarrow$ paradigm shift |
| Breakthrough | Not applicable | Edge-weighting (0 MM $\rightarrow$ 2 MM) |
| Theory | Implicit validation | 42% threshold, geographic dominance |

Table 2: Innovation trajectory comparison showing how panel-driven systematic failure exploration led to breakthrough contributions.

**Critical observation**: User provided the edge-weighting *idea* ("weight edges to preserve minority communities"), but this insight emerged *after* 8+ panel-driven rounds systematically eliminating multi-constraint variations. Traditional work would not have explored alternatives deeply enough to expose the need for paradigm shift.

### 4.5.2 Iteration Depth: 2.0 vs. 5.0 (+150%)

**Traditional iteration**:

- Round 1: User identifies problem, Claude writes methodology
- Round 2: User reviews, suggests edits, Claude revises
- Depth: 1–2 rounds, surface-level revisions
- Driving force: User satisfaction

**Panel-driven iteration**:

- Round 1: Multi-constraint approach $\rightarrow$ Panel identifies limitations
- Round 2: N-way partitioning $\rightarrow$ Panel notes insufficient improvement
- Round 3: Adaptive bisection $\rightarrow$ Panel notes Alabama still fails

- Round 4: 6 tree structure variations → Panel questions fundamental approach
- Round 5: User suggests edge-weighting → Claude implements
- Round 6: Breakthrough (Alabama 0 → 2 MM) → Panel validates
- Round 7: Panel demands comprehensive ablation study
- Round 8+: 140 configurations tested, theory built
- Depth: 8+ rounds with multiple methodological pivots
- Driving force: Panel review satisfaction

**User role shift**: Traditional work—user as director ("write X", "revise Y"). Panel-driven work—user as facilitator (debugging METIS errors, fixing infrastructure) and insight provider (edge-weighting idea at critical juncture). Panel drove scientific trajectory.

### 4.6 Publication Venue Projections

Based on quality assessments, we project target venues and acceptance likelihoods:

| **Corpus** | **Target Venue** | **Acceptance Likelihood** |
|---|---|---|
| Traditional | Regional conferences<br>Workshop papers<br>Domain-specific journals (low-tier) | 60–70% |
| Panel-Driven | Science Advances<br>Nature Computational Science<br>Top political science (APSR, AJPS) | 80–90% (after revision) |

Table 3: Projected publication venues based on quality assessments. Traditional papers target incremental contribution venues; panel-driven work targets top-tier breakthrough venues.

**Reasoning**: Panel-driven work demonstrates (1) novel methodological contribution (edge-weighting paradigm shift), (2) systematic experimental validation (140 configurations), (3) clear theoretical insights (42% threshold, geographic dominance), (4) policy-relevant findings (VRA compliance feasibility), and (5) rigorous comparison to alternatives—characteristics of top-tier computational science and political science publications.

Traditional work demonstrates feasibility and measures outcomes—characteristics of solid regional conference papers or workshop contributions.

## 5 Discussion

Our results demonstrate that panel-driven research achieves +127% quality improvement over traditional Claude-assisted work. We now analyze the *mechanisms* driving this improvement, characterize role dynamics, and discuss implications for AI-assisted research.

### 5.1 Mechanisms Driving Quality Improvement

We identify four mechanisms through which panel review elevates research quality beyond user-directed work.

#### 5.1.1 Mechanism 1: Systematic Questioning

**Panel process**:

Panel reviews systematically ask: "Have you tried alternative approaches? How do you know this is optimal? What if you varied parameter X?"

Example from VRA research:

> *"The multi-constraint approach shows promise, but have you systematically tested alternative partitioning strategies? N-way partitioning might avoid the greedy decisions of recursive bisection."*

This question forced exploration of n-way, adaptive bisection, and 6 tree structure variations, ultimately exposing the fundamental limitation of multi-constraint optimization and motivating the edge-weighting paradigm shift.

**Traditional process**:

User satisfied with working approach. No external pressure to explore alternatives. Alabama's 49.6% result would be noted as a limitation ("this approach doesn't work for all states"), not investigated as a scientific puzzle.

**Impact**: Systematic questioning transforms research from *validation* ("does my approach work?") to *exploration* ("is this the best approach? why or why not?").

#### 5.1.2 Mechanism 2: Embracing Negative Results

**Panel process**:

Panel reviews treat failures as scientific puzzles demanding explanation. Alabama's 49.6% result (0.4 points short of 50% VRA threshold) prompted:

> *"Alabama reaches 49.6%, just 0.4 points short. This suggests a fundamental limit. What geographic factors prevent crossing the 50% threshold?"*

This question led to systematic failure investigation across 6 tree structures, 3 partitioning methods, and constraint relaxation variations. The persistent 49.6% ceiling established a geographic limit, not an algorithmic deficiency, leading to the 42% threshold theory and geographic dominance hypothesis.

**Traditional process**:

Failures are noted as limitations. "Alabama doesn't achieve VRA compliance with this approach" would appear in a limitations section without deeper investigation. Research focus remains on successful states.

**Impact**: Embracing negative results drives theory building. The most important scientific insights often emerge from systematic failure analysis [17], which panel reviews demand but user-directed work rarely prioritizes.

#### 5.1.3 Mechanism 3: Standards Elevation

**Panel process**:

Multi-expert perspectives demand:

- Explicit research questions (not implicit in methodology)
- Enumerated contributions with quantitative claims

- Systematic testing with controls and ablation studies
- Mechanistic explanations (why, not just what)
- Targeted citations supporting specific claims

Claude must meet journal-level standards to satisfy expert review. The bar is "would this pass peer review at a top venue?" not "does the user understand and accept this?"

**Traditional process**:

User satisfaction is the bar. Standards depend on user's familiarity with academic publishing norms. Users without deep research backgrounds may accept narrative structure, general citations, and validation-focused results—all acceptable for accessibility but insufficient for rigorous publication.

**Impact**: Standards elevation enables users without deep research backgrounds to produce rigorous work. The panel provides the expert judgment that the user may lack.

#### 5.1.4 Mechanism 4: Iteration Forcing

**Panel process**:

Review → revise → review → revise cycle continues until panel satisfied. Gate conditions enforce quality thresholds: average score $\geq 2.5/4$, no score $< 2/4$, all P1 items addressed [9]. Research cannot advance until meeting these standards.

VRA research required 8+ rounds:

1. Multi-constraint → insufficient
2. N-way → insufficient improvement
3. Adaptive → still fails
4. 6 trees → fundamental limitation exposed
5. Edge-weighting → breakthrough
6. Validation → panel satisfied
7. Ablation study demanded
8. 140 configs tested, theory built

**Traditional process**:

Draft → user feedback → revise → done. Typically 1–2 rounds. Iteration stops when user is satisfied, which often occurs before expert-level rigor is achieved.

**Impact**: Iteration forcing prevents premature convergence. Many breakthrough contributions require multiple failed approaches to expose fundamental limitations and motivate paradigm shifts. User-directed work rarely iterates deeply enough to reach these insights.

### 5.2 Role Dynamics: Director vs. Facilitator vs. Driver

Our analysis reveals distinct role patterns between traditional and panel-driven research.

### 5.2.1 Traditional Research Roles

**User: Director**

- Identifies research question
- Directs methodology choices
- Reviews drafts and requests revisions
- Makes final decisions on publication
- **Primary driver**: User scientific judgment

**Claude: Executor**

- Executes user directions
- Writes sections as instructed
- Generates figures and tables
- Responds to user feedback
- **Role**: Assistant following user guidance

This mirrors the traditional research assistant model: senior researcher (user) directs, junior researcher (Claude) executes.

### 5.2.2 Panel-Driven Research Roles

**Panel: Scientific Driver**

- Identifies gaps in methodology
- Questions assumptions and demands justification
- Requires systematic testing and alternative exploration
- Validates breakthroughs and demands rigorous evaluation
- **Primary driver**: Panel scientific standards

**Claude: Autonomous Researcher**

- Responds to panel feedback by designing experiments
- Explores alternatives to address panel concerns
- Iterates on approaches until panel satisfaction
- Documents rigorously per panel standards
- **Role**: Researcher responding to peer review

**User: Facilitator + Insight Provider**

- Facilitates execution (debugging, infrastructure fixes)
- Provides domain insights at critical junctures
- Enables Claude to execute panel requirements
- **Role**: Infrastructure provider and domain expert, not scientific director

### 5.2.3 Critical User Contribution: Edge-Weighting Insight

The user's edge-weighting idea ("weight edges to avoid cutting minority communities") appears to contradict the claim that panel drove innovation. However, process tracing reveals:

**Context**: This insight emerged *after*:

- Panel identified multi-constraint limitations (round 1)
- Claude systematically tested alternatives per panel demands (rounds 2–4)
- Alabama remained at 49.6% across all variations (rounds 2–4)
- Panel questioned fundamental approach (round 4)

**User contribution type**: Domain knowledge (minority community preservation) translated into algorithmic strategy (weighted edges).

**Enabling factor**: Panel-driven systematic exploration created the context for this insight. Without 8+ rounds exposing multi-constraint limitations, the need for paradigm shift would not have been apparent.

**Verdict**: Panel was the **primary driver** of scientific investigation, creating conditions for breakthrough. User provided **critical domain insight** at the right moment. This collaboration model—panel driving exploration, user providing domain expertise, Claude executing—may be optimal for AI-assisted research.

## 5.3 Implications for AI-Assisted Research

### 5.3.1 For AI Research Agents

**Autonomy vs. quality tradeoff**: Traditional work positions Claude as executor following user direction (low autonomy, quality depends on user expertise). Panel-driven work positions Claude as autonomous researcher responding to expert review (high autonomy, quality depends on review standards).

**Cognitive load**: Panel-driven work requires Claude to satisfy expert-level scrutiny, design systematic experiments, and build theory—higher cognitive load than executing user instructions. However, this may be optimal for frontier model capabilities [2].

**Research output**: Panel-driven work produces theory building and breakthrough contributions, not just validation and application. This suggests AI research agents can achieve higher scientific impact when guided by structured review rather than user direction alone.

### 5.3.2 For Users

**Reduced burden**: Panel reduces user need to direct every step, identify gaps, and demand systematic testing. Users without deep research backgrounds can produce rigorous work.

**Shifted value**: User value moves from research direction to domain expertise and facilitation. The VRA breakthrough required both panel-driven systematic exploration (rounds 1–4) and user domain insight (edge-weighting, round 5).

**Expertise threshold**: Panel-driven workflows may lower the expertise threshold for producing publication-quality research. However, domain knowledge remains critical at key junctures.

#### 5.3.3 For Research Quality

**Panel as multiplier**: Panel review amplifies research impact from “working methodology” (regional conference) to “publishable science” (top-tier journal). The +127% quality improvement translates to venue tier shifts.

**Systematic peer review goals**: AI-simulated expert review achieves goals of human peer review—identifying gaps, demanding rigor, validating contributions—with temporal efficiency (8 rounds in days vs. months for human review cycles).

**Limitations**: Panel review within a single domain (computational redistricting) with one user limits generalizability. Multi-domain, multi-user validation is needed.

### 5.4 Comparison to Related Work

Our findings complement and extend prior work:

**Revision dynamics** [13]: Showed P1 items account for 72% of score improvement across review rounds. We extend this by showing panel review *generates different types of P1 items* (systematic testing, negative result analysis, theory building) compared to user-directed feedback (clarity, presentation, incremental improvements).

**Hierarchical review architecture** [12]: Showed 37% of PP1 items and 52% of B1 items are emergent patterns not visible at paper level. We extend this by showing panel-driven research elevates *individual paper quality*, not just cross-portfolio synthesis.

**Synthesis methods** [11]: Showed automated P1/P2/P3 classification achieves 94% issue coverage. We extend this by showing P1 items from panel review drive breakthrough innovations (edge-weighting, 42% threshold) that user-directed P1 items do not.

### 5.5 Threats to Validity

#### 5.5.1 Internal Validity

**Sample size**: 3 traditional papers vs. 1 panel-driven paper limits statistical power. However, within-project design controls for domain, author, model, and timeframe, isolating review methodology effects.

**Author bias**: The author served as one rater and produced all papers, introducing potential bias toward panel-driven work. Mitigation: independent second rater, explicit scoring rubrics, process tracing evidence from git history and session notes.

**Confounding factors**: Panel-driven research occurred later (Feb 2026) than traditional work (early 2026), potentially benefiting from accumulated project knowledge. However, the VRA problem was novel (first investigation of VRA compliance vs. general redistricting), suggesting learning effects are minimal.

#### 5.5.2 External Validity

**Domain specificity**: Computational redistricting may not represent all research domains. Replication across domains (NLP, systems, HCI) is needed.

**User expertise**: Single author with software engineering background but limited political science expertise. Users with deeper domain knowledge might achieve higher traditional research quality, narrowing the gap.

**AI model**: All work used Claude Sonnet 4.5 (Jan–Feb 2026). Results may not transfer to models with different capabilities or training.

### 5.5.3 Construct Validity

**Quality dimensions**: Our 10 dimensions emphasize structural rigor and scientific maturity, potentially underweighting accessibility, clarity, and practical impact. Alternative dimension sets might show different effect sizes.

**Scoring subjectivity**: Despite explicit rubrics and dual raters, quality assessment involves judgment. Different raters might score differently, though substantial inter-rater agreement ($\kappa = 0.78$) suggests robustness.

## 5.6 Future Work

**Multi-domain validation**: Replicate analysis across NLP, systems, HCI, and other domains to assess generalizability.

**Multi-user study**: Recruit diverse users (varying expertise levels, domain backgrounds) to test whether panel-driven quality improvements transfer.

**Longitudinal impact**: Track publication outcomes (acceptance rates, citation impact) for traditional vs. panel-driven papers to validate venue projections.

**Mechanism isolation**: Design controlled experiments varying review characteristics (number of reviewers, iteration depth, P1/P2/P3 thresholds) to isolate individual mechanism contributions.

**Cost-benefit analysis**: Quantify time investment for panel review (8+ rounds) vs. traditional work (1–2 rounds) to assess efficiency tradeoffs.

# 6 Conclusion

We presented a comparative analysis of traditional Claude-assisted research and panel-driven methodology using papers from a single computational redistricting project. Our findings demonstrate that structured AI-simulated expert review improves research quality substantially (+127% across 10 dimensions) through four mechanisms: systematic questioning, embracing negative results, standards elevation, and iteration forcing.

## 6.1 Key Findings

**Quality improvement**: Panel-driven research achieves 5.0/5.0 vs. 2.2/5.0 for traditional work. The largest improvements occur in dimensions related to systematic exploration (systematic testing: +400%, negative results: +400%) and scientific rigor (experimental rigor: +150%, hypothesis clarity: +150%, theory building: +150%).

**Innovation trajectory**: Breakthrough contributions (42% demographic threshold, edge-weighting methodology) emerged only through panel-driven systematic exploration of failures. Traditional research would document Alabama's 49.6% result as a limitation; panel-driven research treated it as a puzzle demanding 8+ rounds of systematic investigation.

**Role dynamics**: User role shifts from director (traditional) to facilitator + domain expert (panel-driven). The panel acts as the primary scientific driver, questioning assumptions, demanding systematic testing, and validating breakthroughs—functions typically performed by expert collaborators or advisors.

**Publication impact**: Quality improvements translate to venue tier shifts. Traditional papers target regional conferences (60–70% acceptance likelihood); panel-driven work targets top-tier computational science journals (Science Advances, Nature Computational Science) with 80–90% acceptance likelihood after revision.

## 6.2 Implications

### 6.2.1 For AI Research Agents

Panel-driven workflows demonstrate that AI can function as an autonomous researcher responding to expert review, not merely an assistant executing user directions. This has implications for frontier model capabilities:

- **Higher cognitive load**: Panel-driven work requires satisfying expert-level scrutiny, designing systematic experiments, and building theory. This may be optimal for advanced models [2].
- **Autonomous research capability**: Panel reviews enable AI to conduct investigations, identify gaps, explore alternatives, and build theory—capabilities beyond executing user instructions.
- **Quality control through review**: Structured peer review provides quality assurance for autonomous AI research without requiring user expertise in research methodology.

### 6.2.2 For Human-AI Research Collaboration

Panel-driven workflows offer a new collaboration model balancing autonomy and quality:

- **Reduced user burden**: Users without deep research backgrounds can produce rigorous work through facilitation rather than direction. Panel provides expert judgment.
- **Shifted user value**: User contributions move from research direction to domain expertise and infrastructure facilitation. The VRA breakthrough required both panel-driven systematic exploration and user domain insight at a critical juncture.
- **Lowered expertise threshold**: Panel-driven workflows may democratize research by enabling users with domain knowledge but limited research methodology training to produce publication-quality work.

### 6.2.3 For Research Quality Assurance

AI-simulated expert review achieves goals of human peer review with temporal efficiency:

- **Pre-submission quality control**: Panel review identifies gaps, demands rigor, and validates contributions before venue submission, reducing post-submission revision burden.
- **Systematic improvement**: P1/P2/P3 priority classification focuses revision effort on high-impact issues, with P1 items accounting for 72% of quality improvement [13].
- **Multi-scale synthesis**: Hierarchical review architecture (paper/panel/board tiers) surfaces cross-cutting patterns invisible at individual paper level [12].

## 6.3 Limitations and Future Work

Our study has four primary limitations:

**Sample size**: 3 traditional papers vs. 1 panel-driven paper. Within-project design controls for confounds, but replication across projects and domains is needed.

**Single author**: One researcher produced all papers. Multi-user studies with varying expertise levels would test whether quality improvements transfer.

**Domain specificity**: Computational redistricting may not represent all domains. Replication across NLP, systems, HCI, and other areas is needed.

**Longitudinal validation**: Quality assessments are pre-submission. Tracking publication outcomes (acceptance rates, citation impact) would validate venue projections and quality metrics.

Future work should address these limitations through multi-domain, multi-user longitudinal studies. Additionally, controlled experiments varying review characteristics (reviewer count, iteration depth, P1/P2/P3 thresholds) would isolate individual mechanism contributions, and cost-benefit analysis would quantify efficiency tradeoffs between review paradigms.

### 6.4 Closing Remarks

The central question motivating this work was: **Does structured peer review simulation improve AI-assisted research quality, or merely add process overhead?**

Our evidence strongly supports the former. Panel-driven research achieves +127% quality improvement by transforming the AI's role from executor to autonomous researcher responding to expert review. The panel acts as a scientific driver, systematically questioning assumptions, demanding rigorous testing, embracing negative results, and elevating standards to journal-level rigor.

Critically, breakthrough innovations emerged *because* of panel-driven systematic failure exploration, not despite it. The edge-weighting paradigm shift and 42% threshold theory arose only after 8+ rounds of panel-demanded alternatives exposed fundamental multi-constraint limitations. Traditional research would have documented these limitations and moved on; panel-driven research treated them as puzzles demanding resolution.

This suggests a provocative conclusion: **For AI-assisted research, structured peer review may be more than quality assurance—it may be the primary driver of scientific discovery.** By forcing systematic exploration, embracing failures, and demanding theory building, panel review elevates AI-assisted work from "working methodology" to "publishable science."

As AI research agents become more capable, the question is not whether they can execute user directions (clearly yes), but whether they can conduct autonomous investigations meeting expert standards. Our work suggests the answer is also yes, provided they operate within structured review frameworks that demand rigor, systematic exploration, and theory building.

The future of AI-assisted research may not be better assistants executing user directions, but autonomous researchers responding to expert review—with users contributing domain expertise and facilitation rather than research direction. This collaboration model, tested here on computational redistricting, may generalize to any domain where breakthrough contributions require systematic failure exploration and expert-level scrutiny.

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